Robust Automatic Rodent Brain Extraction Using 3-D Pulse-Coupled Neural Networks (PCNN)

Nigel Chou, Jiarong Wu, Jordan Bai Bingren, Anqi Qiu, and Kai-Hsiang Chuang

***Abstract*—Brain extraction is an important preprocessing step for further processing (e.g., registration and morphometric analysis) of brain MRI data. Due to the operator-dependent and time-consuming nature of manual extraction, automated or semi-automated methods are essential for large-scale studies. Automatic methods are widely available for human brain imaging, but they are not optimized for rodent brains and hence may not perform well. To date, little work has been done on rodent brain extraction. We present an extended pulse-coupled neural network algorithm that operates in 3-D on the entire image volume. We evaluated its performance under varying SNR and resolution and tested this method against the brain-surface extractor (BSE) and a level-set algorithm proposed for mouse brain. The results show that this method outperforms existing methods and is robust under low SNR and with partial volume effects at lower resolutions. Together with the advantage of minimal user intervention, this method will facilitate automatic processing of large-scale rodent brain studies.**

***Index Terms*—MRI, pulse-coupled neural networks (PCNNs), rodent brain, segmentation, skull-stripping.**

I. Introduction

RODENT brains have been used as preclinical models to investigate brain development, disease progression, and neurodegeneration [1]–[4]. With the increasing prevalence of high-resolution MRI studies in rodents to study changes in structure and function, the ability to automatically process rodent brain images has become critical. One essential processing step, brain extraction, is the removal of nonbrain tissue from an image. It is an important prerequisite for further processing such as segmentation, inter-subject registration, and morphometric analysis [5]–[13]. The resulting brain mask can also be used directly to study changes in brain volume [14], [15]. While manual brain extraction by trained individuals provides the most accurate results, it is labor-intensive and operator-dependent, making it unfeasible for high-throughput and large-scale studies, especially ones that utilize high-resolution 3-D imaging. Hence, there is a need for robust and accurate automatic methods to facilitate such studies.

Automatic methods are widely available for brain extraction in humans, utilizing a variety of region-growing [16], edge-based [17], surface-based as well as hybrid [18] approaches. These methods have been validated in numerous human studies [19]–[21]. However, these algorithms often perform poorly on rodent brains due to the different shape and contrasts between brain and surrounding tissues in human and rodent brains. A number of such methods make assumptions of a spherical or ellipsoidal surface [22] or make use of human brain atlases as prior information [23]. Others make use of anisotropic diffusion filtering together with edge-detection [17], but its effectiveness in rodent brains is reduced by the much narrower gap and weaker contrast between the brain and scalp tissue compared to human brains. Due to the failure of existing methods to accurately segment rodent brains, there is a pressing need to develop algorithms that are specifically optimized for rodent brains.

To date, there has been little published work on automatic brain extraction in rodents. In the literature, two methods have been proposed: Constraint level-sets (CLS) [24] and pulse-coupled neural networks (PCNNs) [25]. CLS is a semi-automatic method that uses a level-set algorithm to evolve a user-defined initial surface toward the brain boundary. User intervention is required, involving tracing of the brain boundary on a number of sagittal and coronal slices, which are used both as an initial surface and as constraint points. As with all level-set-based methods, this algorithm is sensitive to its initial surface and parameters.

The PCNN is a biologically inspired neural-network algorithm [26] with numerous applications in image processing, most notably in image segmentation [27]–[29]. Murugavel *et al.* first applied a PCNN algorithm for brain extraction. In their method, a 2-D network was used to segment selected coronal slices of rat head MRI (and hence will be dubbed as 2-D PCNN in the following). It showed improved performance over the Brain-Extraction Tool (BET) [16], a human brain-extraction algorithm. The algorithm's key strengths are its robustness to noise and ability to accurately segment irregular boundaries. However, while this algorithm successfully segments central

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coronal slices, it was unable to correctly identify brain tissue in the anterior slices, where the eyes were larger than the olfactory lobe and in some posterior slices where protruded parts of the brain were isolated. Furthermore, this method requires selection of an optimal brain mask for each slice and hence slows down the throughput especially with high-resolution 3-D image data. Although the authors trained another artificial neural network (ANN) to "automatically" determine the optimal brain mask, the ANN would need to be retrained for different species or for diseased animals. Even with use of the ANN, accuracy would not be consistent across slices and each slice will still need to be checked manually to determine if the correct iteration has been chosen.

In this paper, we extend the 2-D PCNN method proposed by Murugavel *et al.*to overcome the above-mentioned limitations. An important reason for poor performance on certain brain regions was that neighboring information in the third dimension was not considered. We expanded the network to a 3-D model with spherically symmetric 3-D weighting matrices that can be scaled according to the actual voxel dimensions, so that all of the protruding parts across slices can be properly identified. Furthermore, based on the 3-D model, we propose a simple yet effective method to automatically determine the optimal brain mask which does not require a trained ANN nor human intervention, making the 3-D PCNN method fully automatic. The accuracy of the 3-D PCNN method was evaluated on a mouse head MRI acquired using 3-D and 2-D multislice Turbo Spin Echo T2-weighted sequences. Our method was first tested for robustness to noise and partial volume effects by simulating a variety of SNR levels and resolutions. It was then compared with existing human brain-extraction algorithms such as the Brain Surface Extractor (BSE) [17], and rodent brain-extraction techniques: the 2-D PCNN and CLS algorithms.

II. Materials and Methods

A. Overview of Processing Flow

The image was first corrected for bias field using the Nonparametric Intensity Nonuniformity Normalization (N3) algorithm [30]. PCNN was then applied to the entire brain imaging volume, with each iteration including the pixels highlighted by earlier iterations. For each iteration, morphological opening was used to break narrow connections between regions, and the region with the largest volume was selected as the brain mask. Brain mask volume was then plotted against iteration number. This plot was used to select the iteration that best approximates the brain surface. The processing flow is outlined in Fig. 1, and the algorithm is described in detail in the following sections.

B. 3–D PCNN Algorithm

The 3-D PCNN algorithm was expanded based on the 2-D PCNN formulation proposed by Murugavel *et al.* In this implementation, PCNN can be understood as a binary classification algorithm which iteratively groups adjacent pixels with similar intensity. The PCNN algorithm is a unique neural network model based loosely on the visual cortex of cats [26], [27]. It differs from conventional neural networks in that it does not require training, and its distinguishing feature is the addition of a

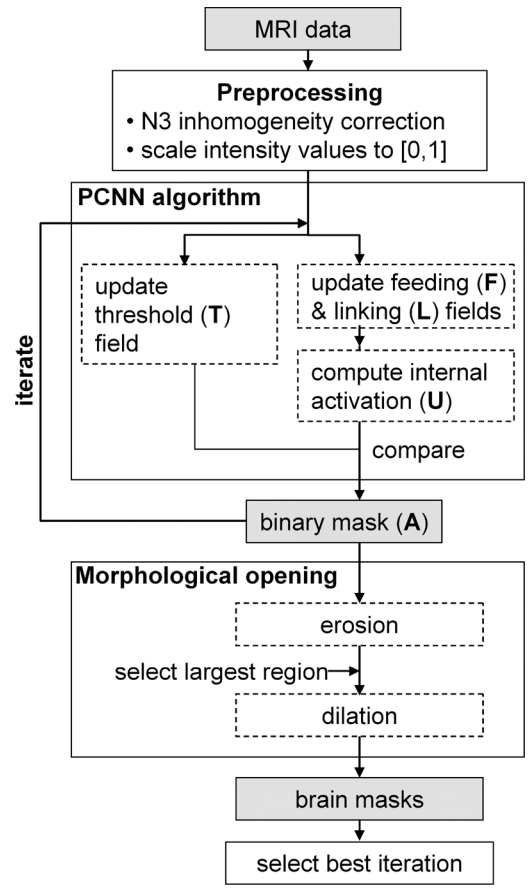

Fig. 1. Processing flow of the 3-D PCNN brain-extraction algorithm.

secondary receptive field, known as the "linking" field, that integrates inputs from adjacent neurons to modulate the primary "feeding" field [26].

Each neuron in the neural network corresponds to one voxel in a grayscale input image. It receives inputs from the voxel itself as well as weighted inputs from surrounding voxels. These inputs are used to compute the "feeding" field, which is updated at each iteration. The "linking" field, which only receives inputs from surrounding voxels, is used to modifying the "feeding" field. The extent of modulation can be controlled by a linkage coefficient. The modified value, known as the "internal activation" of the neuron, is then compared to a "threshold" field, which determines whether the neuron is "switched on." This binary output is then fed into the next iteration.

Neurons are "switched on" more quickly when surrounding voxels have similar intensity values, allowing activation to spread rapidly (within a small number of iterations) throughout regions of similar intensity. However, at regions with high image gradients such as the border between the cerebral cortex and cerebrospinal fluid, it takes many more iterations for adjacent voxels to be "switched on." If a continuous border is present, this will temporarily halt activation of voxels outside the enclosed region until the "feeding" and "linking" fields can exceed the threshold value. As a result, if one records the total number of pixels that have been "switched on" before each iteration, one will find that this value remains relatively

constant for a number of iterations before suddenly increasing. This property is utilized to identify the iteration(s) at which the surface of the mask is at a border.

To explain the algorithm in more detail, let I_{ijk} be the 3-D image matrix, after intensity rescaling to the range of $[0,1]$. F_{ijk}, L_{ijk}, and T_{ijk} representing the "feeding," "linking," and "threshold" fields, respectively, are defined as follows:

$$F_{ijk}[n] = e^{-\alpha_F} F_{ijk}[n-1] + I_{ijk} + V_F(M \otimes Y[n-1])_{ijk} \quad (1)$$

$$L_{ijk}[n] = e^{-\alpha_L} L_{ijk}[n-1] + V_L(W \otimes Y[n-1])_{ijk} \quad (2)$$

$$T_{ijk}[n] = e^{-\alpha_T} T_{ij}[n-1] + V_T Y_{ijk}[n-1] \quad (3)$$

where α_F, α_L, and α_T are decay factors that control how the values of F_{ijk}, L_{ijk}, and T_{ijk} decay over time in the absence of external stimuli. V_F, V_L and V_T are amplification factors that scale the inputs from adjacent pixels to their respective fields. M and W represent 3-D weighting matrices that control the weightage of adjacent voxel inputs. $Y_{ijk}[n-1]$ is the output of the neuron from the previous iteration. The $\otimes$ represents convolution.

All neurons in $\mathbf{F}$ and $\mathbf{L}$ have a value of 0 initially while those in $\mathbf{T}$ have an initial value of 1. For our implementation, $M = W$. The weighting matrix is a normalized 3-D Gaussian function (FWHM $= 0.5$) with peak at the center and radius r, i.e., with a window size of $2r+1$. We chose $r = 3$ as larger values did not improve results and introduced additional computational complexity. In the case of anisotropic voxel dimensions, we scaled the values of the matrix accordingly but retained the dimensions of the window.

The "internal activation" of the neuron U_{ijk} is then computed from F_{ijk} and L_{ijk} as

$$U_{ijk}[n] = F_{ijk}[n](1 + \beta L_{ij}[n]) \quad (4)$$

where β is the linkage coefficient controlling the effect of the "linking" field on U_{ijk}. Y_{ijk}, the output of the neuron, is obtained by comparing the threshold value $T_{ijk}[n]$ to the internal activation $U_{ijk}[n]$ as

$$Y_{ijk}[n] = \begin{cases} 1, & \text{if } U_{ijk}[n] > T_{ijk}[n] \\ 0, & \text{if } U_{ijk}[n] \le T_{ijk}[n]. \end{cases} \quad (5)$$

To ensure that any voxel that is switched "on" remains "on" even if it falls below the threshold value in later iterations, an accumulated output A_{ijk} is then obtained by accumulating all pixels before and including the present iteration:

$$A_{ijk}[n] = A_{ijk}[n-1] \cup Y_{ijk}[n]. \quad (6)$$

This binary output A_{ijk} is then modified further using morphological operations to identify the brain mask. The parameters used in our PCNN implementation were similar to [26] and [25] and are summarized in Table I. Variation of these parameters did not appear to affect accuracy of the final brain mask, although it slightly modified the shape of the iteration plot.

TABLE I
PARAMETERS USED IN PCNN

PCNN constant	value	interpretation
τ_F	0.3	Feeding decay time-constant
τ_L	1	Linking decay time-constant
τ_T	10	Threshold decay time-constant
V_F	0.01	Feeding amplification factor
V_L	0.2	Linking amplification factor
V_T	20	Threshold amplification factor
β	0.2	Linking coefficient
r	3	Linking weightage field radius

$\alpha_{(F,L,T)} = ln2/\tau_{(F,L,T)}$.

C. Morphological Operations

For each iteration, morphological erosion is performed on the resultant mask $\mathbf{Y}$, using a spherical structural element S_{2p+1} of radius p and dimension $2p+1$:

$$S_{2p+1} = \begin{cases} 1, & \text{if } \sqrt{(i-(p+1))^2+(j-(p+1))^2+(k-(p+1))^2} \le p \\ 0, & \text{if } \sqrt{(i-(p+1))^2+(j-(p+1))^2+(k-(p+1))^2} > p \end{cases}. \quad (7)$$

This removes the narrow bridges connecting large continuous regions and allows for isolation of the brain mask from adjacent structures. The spherical shape of the structural element was chosen to give the smoothest mask, and p was chosen to be 4 for best results on our data.

In order to accommodate anisotropic voxel resolutions, for instance, in 2-D multislice datasets where the slice thickness is greater than the pixel dimensions of the slice, the structural elements (S) were scaled proportionally. For example, in our 2-D multislice dataset, we use an ellipsoidal structural element to account for the larger voxel dimension in the through-plane direction.

Following the assumption that the brain is the largest continuous structure in the field of view, we choose the largest volume in each iteration as the brain mask. After the brain mask is chosen, a morphological dilation operation is performed with the same structural element to restore the size of the brain mask.

D. Selection of Iteration With Optimal Brain Mask

A plot was made of brain mask volume against iteration number (Fig. 2). The optimal brain mask can be found within a "plateau" region of the plot, usually in the center of the plateau. The optimal iterations can be determined manually by inspection of the plot. In all our datasets, the iteration plot displayed a clear "plateau" region, allowing for easy identification of the best brain masks. The iterations stop when the volume of the brain mask exceeds a threshold of 800 mm^3 for mouse brains. The threshold value can be further reduced to cut down computation time or increased in the case of rat brain extraction.

An automatic method for determining the optimal iteration was tested and compared with manual selection. In this method, we first limit the iterations to be searched within a range appropriate for the brain size of the animal. For example, we used

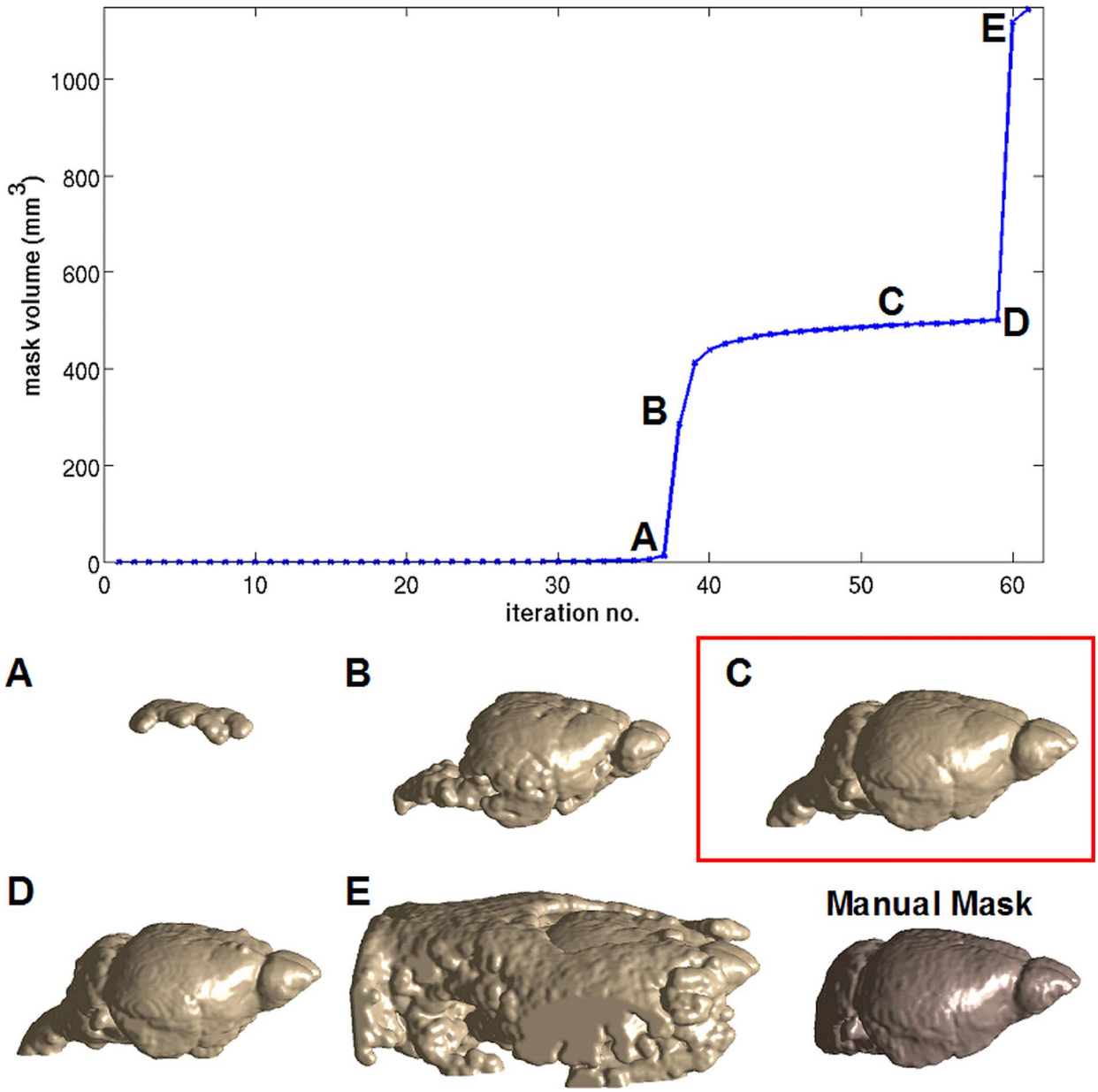

Fig. 2. Illustration of the generated brain mask volume at each iteration in 3-D PCNN showing brain masks at five selected iterations. (a) Iteration 37. (b) Iteration 38. (c) Iteration 52. (d) Iteration 59. (e) Iteration 60. A Manual mask (not including brain-stem) is shown at the lower right corner as a reference. Iteration 52 is the best iteration. Note that iteration 59, though not the ideal brain mask, still conforms relatively well to the manual brain boundary.

the range of 100–550 mm^3 for our mouse brain data. We then find the largest decrease and increase in gradient over this region, which correspond approximately to the start and end of the "plateau" region, respectively. The ideal iteration is chosen as the iteration halfway between these two iterations.

E. MRI

All animal procedures were approved by the Institutional Animal Care and Use Committees of the Biomedical Sciences Institutes, Agency for Science, Technology and Research (A*STAR). Ten adult male C57BL/6 mice (23–28 g) were initially anesthetized by 1.5% isoflurane mixed in 2:1 air:O_2. The anesthetized animals were secured in a stereotactic holder with breathing and temperature monitored during imaging (SAII, NY, USA). The Isoflurane level was varied between 1%–1.5% to maintain a stable breathing rate of 60–90 breaths per minute. Body temperature was maintained using a water heating system.

Images were acquired on a Bruker 7T/20 cm ClinScan (Bruker BioSpin MRI GmbH, Ettlingen, Germany) using a volume coil for transmit and a Bruker mouse brain 2×2 array coil for receive. 2-D T2-weighted Turbo Spin Echo (TSE T2-wt) images were acquired in five mice with the following parameters: TR/ TE $=$ 2710/42 ms, echo train length $=$ 37. The field of view (FOV) was 25 mm $\times$ 25 mm, with a matrix of 256×256 (pixel size $= 98\ \mu$m $\times$ 98 μm). 51 coronal slices of 0.30 mm with no slice gap were acquired to cover the whole brain (total acquisition time $=$ 10 min).

The other five mice were scanned with a 3-D TSE T2-wt sequence. Acquisition parameters were TR/TE = 2000/46 ms, echo train length = 13 and total acquisition time = 1 hr 23 min. The field of view (FOV) was 9 mm × 13 mm × 25 mm, with a matrix of 88 × 140 × 256 (voxel size = 100 μm × 98 μm × 98 μm).

F. Implementation and Evaluation

The 3-D PCNN algorithm was implemented in MATLAB R2008b (Mathworks, Natick, MA, USA). 3-D visualization of brain surface, simulation and evaluation were also carried out in MATLAB. Inhomogeneity correction was performed using the N3 algorithm in MIPAV 4.0.3 (National Institutes of Health, Bethesda, MD, USA), with field distance = 2.5 mm, maximum iterations = 300, end tolerance = 0.001 and kernel FWHM = 0.15. 3-D PCNN was also implemented in C++ and compiled with cmake v2.8.1, but evaluation of this implementation is not included in this paper.

The 3-D PCNN brain-extraction algorithm was first tested for its robustness to noise and partial volume effects by comparing results at different simulated noise levels and resolutions respectively. It was then evaluated both quantitatively and qualitatively against three other automatic methods: BSE, CLS, and the original 2-D PCNN algorithm [25].

To simulate varying noise levels, Gaussian noise was added to both 2-D multislice and 3-D datasets such that the SNR, defined as a ratio of mean intensity in cortex to standard deviation of background, was 10 and 5 for each image. SNR was measured on rectangular ROI drawn manually on both cortex and background over ten slices in each dataset. Results were then compared to the original images, where SNR ranged from 32 to 44 for the 3-D dataset and 21 to 37 for the 2-D multislice dataset.

To simulate the partial-volume effects at lower resolutions, we downsampled the 3-D dataset by 2 × and 3 × by averaging adjacent pixels in all three dimensions. Downsampling was first done on the plane of each coronal slice, with both x and y directions downsampled equally. Downsampling was then applied in the direction perpendicular to the coronal plane (through-plane direction) to simulate the effect of using larger slice-thickness in 2-D multislice acquisitions.

3-D PCNN was compared to three other automated methods: CLS, 2-D PCNN and BSE. The first two methods were designed for rodent brains, while the third (BSE) was developed for human brains. Another human brain-extraction algorithm, the BET as implemented in MIPAV, was tested but not included in this comparison as its performance on mouse brain data was poor. It consistently excluded large sections of brain tissue, despite attempts at adjusting parameter settings.

BSE combines anisotropic diffusion filtering with edge-detection, followed by morphological operations to identify the brain mask. BSE was applied to the entire image volume using BrainSuite09 (Laboratory of NeuroImaging, UCLA, CA), with the following parameters: diffusion iterations = 3, diffusion constant = 25, edge constant = 0.064, erosion size = 1 pixel.

The CLS technique is based on a variation of the level-set method [31] with the incorporation of prior knowledge as constraint points. The technique works by evolving an initial level surface iteratively towards the brain boundary, where the image gradient is the highest. This is controlled by the weighting factors μ, λ, and ν. μ determines the amount of weight given to the internal force which penalizes deviation of the level set function from the previous iteration, while λ and ν determine the weight given to the driving force. The user must first manually extract the brain region on three sagittal and two axial slices to serve as prior knowledge and define the initial level surface. Constraint points are then extracted from these orthogonal contours in the coronal view. The initial level surface is defined by joining the constraint points with straight lines in every coronal slice, then stacking these contours to form a closed surface. In addition, by increasing μ and decreasing λ and ν at the constraint points, the propagation of the level surface is blocked from going beyond these points. Uberti *et al.* implemented CLS on entire image volumes (3-D CLS) and separately on each slice (2-D CLS). We chose to use 3-D CLS as it performed better in T2-weighted images. 3-D CLS was applied to the 2-D or 3-D image volumes with weighting factors $\mu = 0.04$, $\lambda = 5.0$, and $\nu = -3$, while those of the constraint points were set to $\mu = 0.05$, $\lambda = 0.0$, and $\nu = 1$. CLS was implemented in MATLAB using an interface obtained from the authors [24].

To quantitatively evaluate the effectiveness of 3-D PCNN and other automatic methods, manual masks were traced separately by two raters on all datasets with reference to the Paxinos and Franklin atlas [32] and used as the gold standard. Masks were drawn on coronal slices, starting from the slice where the cerebellum is first clearly visible. These manual brain masks were outlined using the volume of interest (VOI) tool in MIPAV. In all cases, three indexes were used to measure similarity of the brain masks from each automatic method ($\mathrm{M_{auto}}$) to the manual standard ($\mathrm{M_{manual}}$).

Jaccard index:

$$\text{Jaccard index} = \frac{|M_{\text{auto}} \cap M_{\text{manual}}|}{|M_{\text{auto}} \cup M_{\text{manual}}|}. \tag{8}$$

True-positive rate (TPR):

$$\text{TPR} = \frac{|M_{\text{auto}} \cap M_{\text{manual}}|}{|M_{\text{manual}}|}. \tag{9}$$

False-positive rate (FPR):

$$\text{FPR} = \frac{|M_{\text{auto}} - M_{\text{auto}} \cap M_{\text{manual}}|}{|V - M_{\text{manual}}|} \tag{10}$$

where V is the total head volume that includes all voxels greater than 5% of the maximum intensity. The Jaccard index is a measure of how closely the automatic mask overlaps the manual mask, and is used as the main index for comparison. The TPR measures the fraction of pixels correctly labeled by the automatic mask, while the FPR gives an indication of how many pixels identified by the automatic mask were outside the manual mask.

The Jaccard index of the manual masks between the two raters was computed, and the mean Jaccard index used as a measure of inter-operator variability. The Jaccard index, TPR and FPR were computed for each automatic mask relative to both sets of manual masks. The two values were then averaged, and used to

TABLE II
SIMULATION OF DIFFERENT SNR CONDITIONS

SNR (dataset)	Jaccard Index		TPR		FPR	
	Mean	SD	Mean	SD	Mean	SD
Original SNR (SNR = 32-44, 3D Dataset)	0.960	0.002	0.981	0.002	0.030	0.008
SNR = 10 (3D dataset)	0.959	0.002	0.982	0.003	0.027	0.006
SNR = 5 (3D dataset)	0.961	0.005	0.984	0.002	0.028	0.009
Original SNR (SNR = 21-37, 2D Dataset)	0.948	0.005	0.976	0.003	0.025	0.010
SNR = 10 (2D dataset)	0.944	0.008	0.976	0.005	0.027	0.009
SNR = 5 (2D dataset)	0.933‡	0.010	0.974	0.008	0.030‡	0.005

‡ indicates significant deviation of low SNR simulation from the original dataset ($p < 0.05$ using Student's T-test)

TABLE III
SIMULATION OF DIFFERENT RESOLUTIONS ON 3-D DATASET

Downsampling direction (Voxel dimensions)	Jaccard Index		TPR		FPR	
	Mean	SD	Mean	SD	Mean	SD
Original resolution (vox = $100 \times 100 \times 100\mu m$)	0.960	0.002	0.981	0.002	0.030	0.008
Through-plane (vox = $100 \times 100 \times 200\mu m$)	0.959	0.002	0.983	0.002	0.017‡	0.009
Through-plane (vox = $100 \times 100 \times 300\mu m$)	0.960	0.002	0.978	0.001	0.014‡	0.003
In-plane (vox = $200 \times 200 \times 100\mu m$)	0.957‡	0.002	0.981	0.003	0.024‡	0.006
In-plane (vox = $300 \times 300 \times 100\mu m$)	0.948	0.011	0.966	0.004	0.014‡	0.002

‡ indicates significant deviation of low-resolution simulation from the original dataset ($p < 0.05$ using Student's T-test)

compute the mean and standard deviation (SD) over the five animals in each dataset. To test for significant difference between the automatic methods, we used a paired Student's T-test and regarded p-values below 0.05 as significant.

For the 2-D PCNN case, the iteration with the best brain mask was determined manually on each slice as the iteration plot was inconsistent and often did not show a well-defined plateau region. This method was only tested on the 2-D multislice dataset due to its smaller number of coronal slices and the need for manual inspection of iterations for each slice. On slices where nonbrain structures appeared to be the best iteration (e.g., slices in olfactory bulb region), no mask was selected for that slice. For 3-D PCNN, all images produced clearly defined iteration plots, with multiple iterations in the plateau region giving good brain masks. Hence, the iteration with the mask closest to the manual mask was used for comparison.

III. RESULTS

A. Inter-Operator and Intra-Operator Variability

To ensure the reliability of the manual gold-standard, we used the Jaccard index to evaluate the similarity between the masks outlined by the two raters. For the 3-D dataset, the Jaccard index was 0.963 with $\mathrm{SD} = 0.005$. For the 2-D multislice dataset, the index was 0.948 with $\mathrm{SD} = 0.004$. One of the raters repeated drawing of manual masks a second time. The Jaccard index of these two masks relative to each other was 0.959 for the 3-D dataset and 0.959 for the 2-D multislice dataset.

B. Performance at Different Simulated SNR Levels

Table II summarizes the Jaccard index, TPR, and FPR obtained when comparing the mouse brain identified by 3-D PCNN at varying SNRs to the manual mask. For the 3-D TSE T2-wt dataset, there was almost no change in Jaccard index, TPR, and FPR (for all comparisons, $p > 0.16$). For the 2-D TSE T2-wt dataset, when SNR was reduced to 5, the mean Jaccard index showed a significant decrease of about 0.015 ($p = 0.005$) while the FPR increased by 0.005 ($p = 0.01$). The True-positive rate did not change significantly.

C. Performance on Downsampled Images

To evaluate the performance under low-resolution conditions where partial volume effects may degrade accuracy, we downsampled the images in both the coronal plane and through-plane directions. Table III lists the Jaccard index, TPR, and FPR for brain extraction at varying resolutions both in-plane and through-plane. When in-plane resolution of each coronal slice was decreased isotropically, the Jaccard index decreased by 0.012 when it was downsampled $3\times$. Both TPR and FPR decreased, with FPR being significant ($p < 0.007$). When resolution was decreased $3\times$ in the through-plane direction (at $3\times$ downsampling the resolution is identical to that of the 2-D dataset) there was no

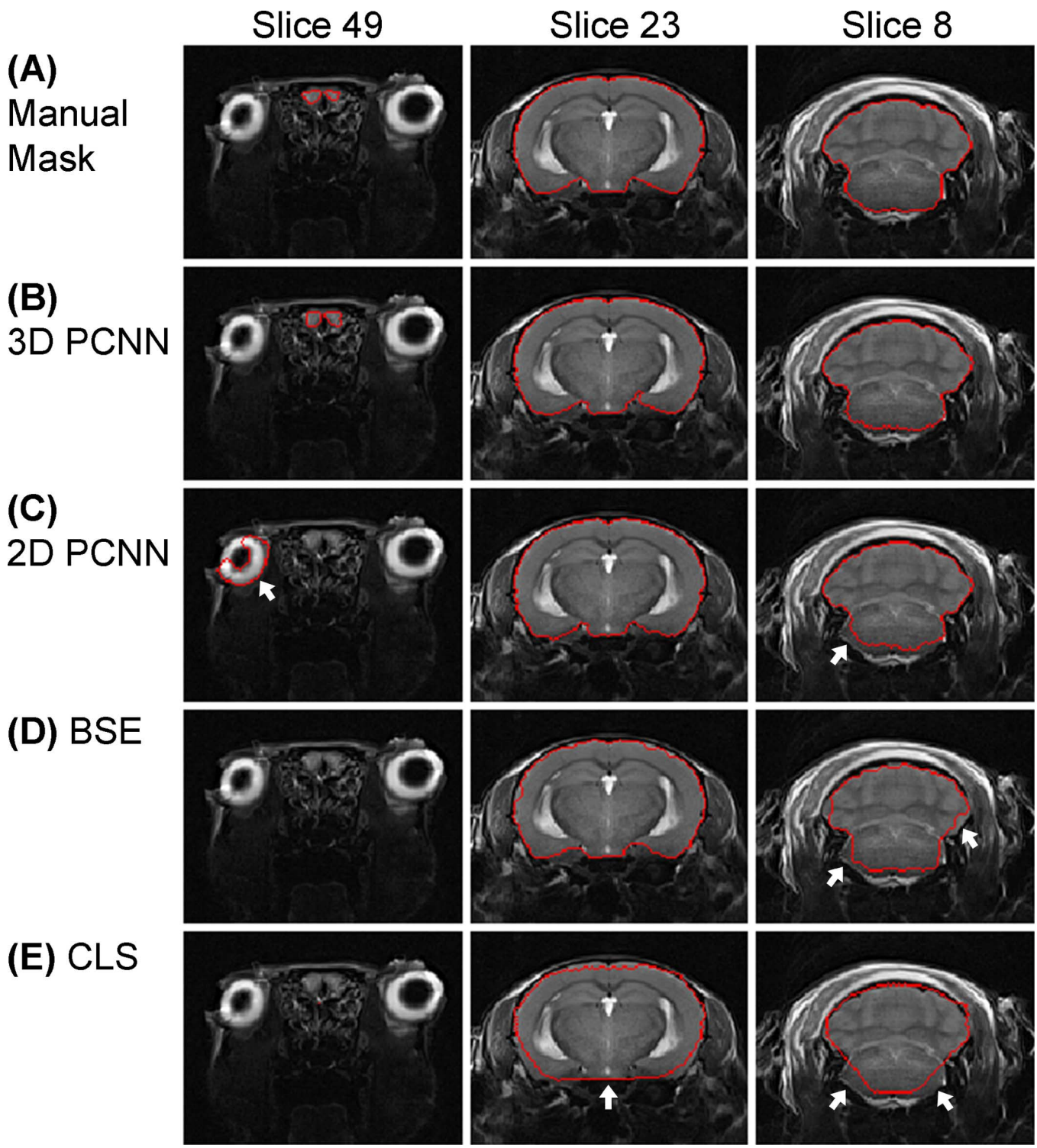

Fig. 3. Comparison of brain masks obtained using various automatic methods on selected slices of one mouse brain acquired with 2-D TSE T2-wt sequence. From left to right are slices covering the following structures: Olfactory bulb; Hypothalamus, thalamus and cerebral cortex; Cerebellum and brainstem. (A) Manual mask (B) 3-D PCNN (C) 2-D PCNN (D) BSE (E) CLS algorithm. Arrows indicate regions having large deviations from the true boundary.

significant change in the Jaccard index ($p > 0.14$) but there was significant decrease in FPR ($p < 0.02$).

D. Comparison With Other Automatic Methods

Fig. 3 shows examples of the brain boundaries identified by the four methods on three slices of a 2-D multislice dataset. Of all of the automatic methods, only 3-D PCNN successfully segments the olfactory bulb. 2-D PCNN incorrectly identified one of the eyes as the brain mask while CLS and BSE are unable to detect any brain tissue in this slice. For coronal slices in the middle of the brain, both 3-D PCNN and 2-D PCNN perform equally well, with BSE having slight errors and CLS

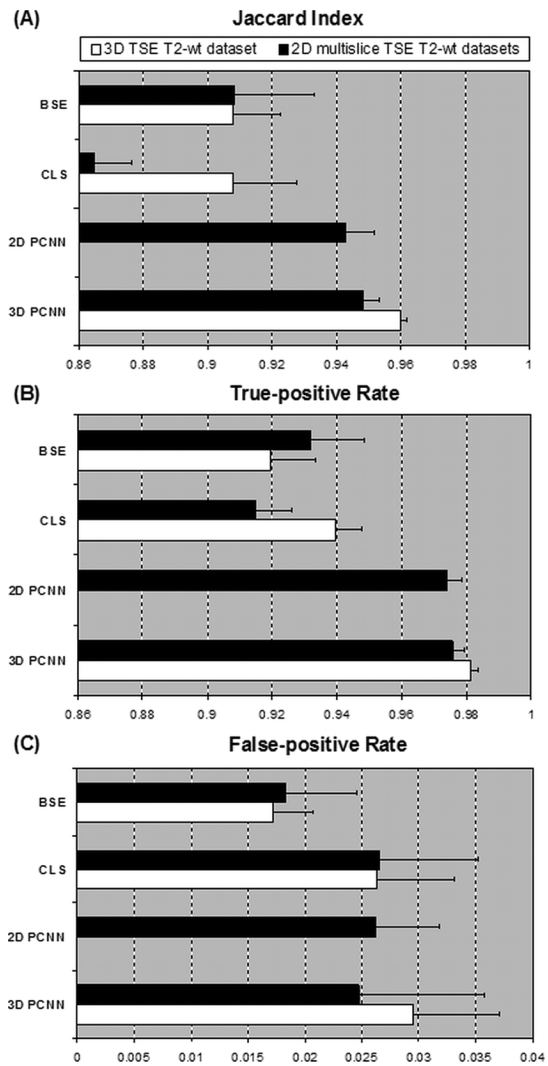

Fig. 4. Comparison of accuracy of various automatic methods using (a) Jaccard index, (b) TPR, and (c) FPR.

unable to correctly segment the lower surface of the brain. For slices showing cerebellum and brainstem, 3-D PCNN performs slightly better than 2-D PCNN in the lower contrast areas of the brain stem. BSE and CLS perform relatively poorly on these slices.

A quantitative comparison is given in Fig. 4. In both datasets, 3-D PCNN performed better than BSE and CLS, with significantly higher Jaccard index and TPR than BSE ($p < 0.02$) and CLS ($p < 0.003$). In all cases except one, 3-D PCNN performed better than 2-D PCNN in Jaccard index, although the difference was not statistically significant ($p > 0.31$). The FPR was comparable across 3-D PCNN, 2-D PCNN and CLS, but slightly lower in BSE ($p = 0.6$ for 3-D dataset and $p = 0.02$ for 2-D multislice dataset) because BSE tends to underestimate the brain mask.

IV. Discussion

We have shown that 3-D PCNN is robust to noise and partial-volume effects. It outperforms 2-D PCNN, the CLS algorithm as well as BSE in all datasets tested. One reason for the improved performance of 3-D PCNN is that it takes both spatial and intensity information into account simultaneously and does not make any assumption of shape. This property makes it less sensitive to small intensity variations, and hence robust to noise, while being able to accommodate the irregular contours and protrusions of the rodent brain. Both factors are critical for segmenting rodent brain MRI volumes. Especially, the expansion of the typical 2-D network model into 3-D with a proportional weighting function allows spatial continuity in all dimensions to be taken into account more accurately. The morphological operations were improved by utilizing structural elements that produce a smoother boundary and including a morphological dilation step after selection of the brain mask to restore the original volume of the brain. In addition, by exploiting the clear plateau period in the iterations of this 3-D model and prior knowledge of the expected brain volume, the brain mask can be automatically determined without unnecessary iterations and human intervention. This also reduces the overall computation time of the 3-D network.

A. Robustness of 3-D PCNN

Tests at different SNR levels show that 3-D PCNN is extremely robust to noise and has minimal (2-D multislice dataset) or no (3-D dataset) reduction in accuracy even at very high noise levels. It also works relatively well when images were downsampled both in the coronal-slice plane and in the through-plane direction, showing minimal degradation of performance in our simulations.

3-D PCNN is also robust with respect to the selection of iterations. In most of the mouse brain volumes tested, there were a number of iterations (up to 12 in the 3-D dataset) in the "plateau" region with similar volume that all approximate the reference brain mask relatively well. Hence, if the iteration chosen deviates from the optimal iteration by a few iterations, a good brain mask will still be obtained.

B. 3-D PCNN Versus 2-D PCNN

3-D PCNN offers improvement over 2-D PCNN in terms of TPR and Jaccard index. It has the added advantage that optimal extraction involves only one iteration selection for the entire volume. In 2-D PCNN, an individual selection is required for every slice. The reduced user intervention is especially important for processing high-resolution 3-D imaging data. Furthermore, in all but the slices in the middle of the brain, the iteration plot produced by 2-D PCNN did not show a clear "plateau" region, making it difficult to determine the correct iteration, whether manually or using automatic methods. In contrast, 3-D PCNN in all cases gave a well defined "plateau" similar to Fig. 2, which is an indication that a boundary has been reached, allowing for accurate selection of the optimal iteration.

Automatic methods for determining the optimal brain mask on a coronal slice, such as the ANN used by Murugavel *et al.*, may help to reduce user intervention but do not overcome the inherent problems in doing segmentation on individual slices rather than whole volumes. For example, each slice must still be manually checked for the correct iteration. This is especially so in the anterior and posterior coronal slices where the "plateau"

region is poorly defined and any method to automatically determine the optimal iteration is likely to fail. In addition, using an ANN requires training, which will have to be repeated on a number of animals whenever a new strain or disease model is used. These limitations make 2-D PCNN cumbersome to use in large-scale studies.

Another important benefit of 3-D PCNN is that it allows knowledge of brain volume (e.g., 450–550 mm^3 for adult mice) to be incorporated in the algorithm, which was not possible in 2-D PCNN, where brain area varies from slice to slice. Brain volume information can be used to aid selection of the correct iteration by limiting the range of iterations to check. It can also be used to significantly cut down computation time by stopping iterations once the brain mask volume exceeds the maximum expected brain volume.

In our experiments, 3-D PCNN always produced iteration plots with clearly defined plateau regions. This property, together with our knowledge of expected brain volume allowed us to devise a simple but effective way to automatically determine the optimal iteration. We compared the average Jaccard index for each dataset from iterations selected with this method to that obtained from iterations with the highest Jaccard index. The difference in Jaccard index was smaller than 0.002 for both 3-D and 2-D multislice datasets. Hence, this would be a viable method to aid in the selection of the optimal iteration.

Although 3-D PCNN inevitably has a higher computational complexity due to whole volume processing, it is still preferable to 2-D PCNN due to its consistent performance, use of out-of-slice information, ease of selection of optimal brain mask and minimal requirement for user intervention. Computation time on a PC with Intel Xeon X5450 3.0 GHz and 16 GB RAM running Linux without use of parallel processing was 32–39 min for the 2-D multislice dataset and 38–48 min for the 3-D dataset using 3-D PCNN, compared with 13 min on the 3-D dataset and 7.5 min for the 2-D multislice dataset using 2-D PCNN. Note that the code used in our evaluation has not been fully optimized, hence computation time can be further decreased for example by using Fourier convolution for the convolution steps or reducing the brain mask volume at which to stop iterations. Moreover, a significant portion of the computation time was taken up by the morphological operations (the main PCNN algorithm took up comparatively little time), hence utilizing more efficient code for the morphological operations may also greatly improve computational efficiency.

C. Comparison With Other Automatic Methods

Among human brain-extraction methods, BSE appears to be one that is better suited for rodent brains due to its minimal use of human brain specific assumptions. However, it does not perform as well in rodent brains due to the smaller number of pixels and reduced contrast between the brain surface and adjacent tissue. In many cases, BSE detects edges slightly inside the surface of the brain and erroneously designates these edges as part of the brain surface. This leads to reduced performance as well as a strong tendency to underestimate the brain surface, which accounts for its lower FPR relative to all other methods tested.

In our comparison, CLS performed much worse than 3-D PCNN and 2-D PCNN. It showed similar results to BSE in the 3-D dataset but had even worse performance than BSE in the 2-D multislice dataset, despite being optimized for rodent (mouse) brains. One reason for this poor performance could be that the image resolution of 100 μm used in our experiment is lower than the 78 μm used in Uberti's paper [24]. As a result, the partial volume effect along the brain/nonbrain boundary for the images used in our experiment may result in reduced accuracy compared with that of Uberti. CLS is also strongly dependent on initial conditions and choice of landmarks which may contribute to its inability to correctly converge on the true brain surface. Finally, the need for a regularization term (μ) prevents the algorithm from correctly detecting the irregular contours especially at the ventral part of the brain where curvature changes rapidly (see Fig. 3). In our experiment, even with the incorporation of manually segmented orthogonal slices as prior knowledge, the algorithm is unable to converge to the brain boundary well. Increasing the number of manually segmented sagittal slices from 5 to 7 did not improve the results of CLS brain extraction. Adding more landmarks where there are larger curvature changes may help to improve the problem, but will significantly increase user intervention.

D. Limitations of 3-D PCNN

3-D PCNN does have some limitations. Like all of the other automatic methods mentioned in this paper, it has difficulty segmenting images with low contrast between cortex and skull, such as in T1-weighted rodent brain images acquired at high field. We also acquired 3-D MPRAGE T1-weighted images in some animals. However, due to the low intensity of gray matter (as opposed to T2-weighted images where gray matter has high intensity), the contrast of the brain boundary was rather low. This may be why most existing MRI rodent brain atlases were acquired with T2-weighting [33]–[37]. Therefore, it is recommended to acquire T2-weighted data for segmentation and registration.

V. Conclusion

We have shown that 3-D PCNN is an effective and robust method that is well suited for rodent brain extraction. It performs better than existing human and rodent brain extraction algorithms, while overcoming the issues faced by the 2-D PCNN method. This method will facilitate application of automatic registration and morphometric analysis methods to rodent brains.[1]

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[1]Software for 3-D PCNN brain extraction will be freely available on our websites. The MATLAB and C++ executables will be available at http://www.sbic.a-star.edu.sg/research/lmi/software.php and http://www.bioeng.nus.edu.sg/cfa/mousebrainatlas/index.html.

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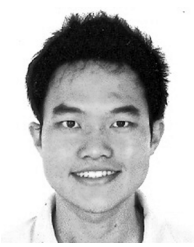

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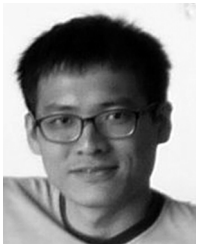

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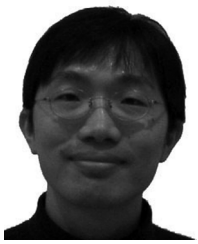

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